

Frank Yang Dininno

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University of California, San Diego — B.S. Nanoengineering (Expected Jun 2026)

GPA: 3.79/4.00

Technical Skills

Software: Python, Java, Bash, Linux, Conda, Pandas, NumPy, serial communication, open-source FEM & FDTD simulation tools (FEMWELL, MEEP), Lumerical, Quantum Espresso, Biochemical tools (RFdiffusion, Addgene, ChimeraX), CAD software (Blender, Autodesk Inventor, Onshape, Klayton, OpenSCAD, Gmsh).

Cleanroom: Nano3 certified and trained.

Metrology: Ellipsometer, SEM, AFM, optical microscopy, stylus profiler, thin-film stress monitor.

Deposition: PECVD, spin coating. **Etching:** RIE, SLR (plasma etchers). **Photolithography:** MLA (Heidelberg), photoresist development.

Shop: Milling machine, resin 3D printing, FDM 3D printing, general machining and fabrication, laser system alignment.

Languages: Mandarin, English

Projects

Laser Annealing of Silicon-Rich Nitride for Waveguide Creation — *Primary Researcher*. Simulating and experimentally studying refractive index modification in Si-rich nitride thin films. Uses MEEP for beam propagation and power absorption, an FDTD thermal simulator for heat distribution, Pandas/NumPy for data fitting, and FEMWELL for mode calculations. Physical setup includes ESP300-controlled stages and a 529 nm high-power laser.

Nano-Innovator-Grant Proposal: Nanosecond Switchable Langmuir Nanopores — *Sole Drafter*. Theorized and proposed a photochemical-based nanopore array (included documentation in website portfolio) that introduced chemical headgroups into a solvent with nanosecond optical switching speeds.

Node Form (Blender Add-on) — *Sole Developer*. Software extension for Blender enabling mathematical animation generation through node-based procedural design. <https://github.com/FyDininno/NodeForm>

Blender Finite Element Mesh Generator/Exporter — *Sole Developer*. Creates finite element meshes directly within Blender and exports to simulation-ready formats.

Graphene Compression Research (Senior Project) — *Researcher*. Designed and manufactured a two-tonne compression device in the UCSD metal shop to study graphene composites for thermal interface materials; achieved improved thermal conductivity in compressed structures.

Elastomeric Graphene Thermal-Interface Materials Research — *Primary Researcher*. Working to develop a hybrid elastomeric, vertically-aligned graphene composite for computer chip heat dissipation and ease of installation. Purchased machinery and developed sophisticated chemical process for industrial scale-up.

Eagle Project — *Designer & Builder*. Constructed a mobile caged garden bed system (included documentation in website portfolio); managed design, materials, and build process.

Work Experience

Silicon Nitride Waveguide Fabrication and Characterization — *Coworker*. Helped develop silicon nitride etch recipes alongside university graduate students and performed waveguide characterization using metrology tools and software.

Classes Taken

Electromagnetic Optics; Optical Information Processing & Holography; Laser Electronics; Waves, Thermodynamics, Optics (PHYS 2C); Relativity & Quantum Physics; Electronic Devices & Circuits; Materials Science; Thermodynamics of Materials; Micro/ Nanofabrication; Nano Systems Characterization; Polymeric Materials; Photolithography & Microfabrication; Density Functional Theory; Heat Transport; Physical Chemistry; Solid State Physics; Biochemical Principles; Crystallography; Organic Chemistry; MATLAB for Engineers; Nanomaterials Synthesis Lab; General Chemistry Lab; Differential Equations; Nuclear Physics Principles; Honors Calculus (31AH/BH/CH); Abstract Algebra. *Enrolled:* Extremal Combinatorics/Graph Theory, Solid State Devices, Systems and Signals, Introduction to Deep Learning and Applications

Honors & Leadership

Achieved the rank of **Eagle Scout** (Nov 2021, Troop 730). **FIRST Robotics CAD Lead** – 3256 Warriorborgs (2021 World's Advancement). Listed contributor of *Geometric Algebra for Electrical Engineers* by Peter Joot (<https://peeterjoot.com/archives/math2025/GAelectrodynamics.color.V0.3.6.4.pdf>).